# A Bayesian Approach to Merger Review

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## 1 Introduction

Merger review is inherently a problem of inference under severe data constraints. Antitrust agencies and practitioners are often asked to evaluate the competitive effects of a proposed transaction using a single cross-section of market shares, prices, and sometimes margin information. In this setting, standard econometric approaches that rely on large-sample variation are not available. Instead, practitioners typically impose a structural model of competition, most often Bertrand pricing, and use that model to simulate post-merger outcomes.

That reliance on a single conduct assumption is fragile. Prior work shows that merger predictions can vary sharply across standard models of firm behavior, including Bertrand, Cournot, and auction-based competition (Panhans and Taragin 2023). Related work on screening thresholds shows that this model dependence can translate into instability in concentration-based policy rules. The core problem is therefore not merely that merger simulation is sensitive to assumptions. It is that the economic conclusions of a merger review can depend critically on assumptions about firm conduct that are usually untested, and often untestable, in the information set available to agencies.

This paper provides a framework for conducting merger analysis under such **model uncertainty**. Rather than selecting a single model of competition, we treat conduct as an object of inference. We embed multiple canonical models–Bertrand, Cournot, second-score auction, and monopolistic competition–within a unified structural framework and use Bayesian methods to estimate their posterior probabilities given the observed data. Merger effects are then computed as probability-weighted averages across models, so that policy conclusions reflect uncertainty about the underlying competitive environment.

The novelty is not Bayesian model averaging in the abstract. Bayesian Model Averaging (BMA) is a standard statistical tool. Our contribution is to apply it to **structural conduct models**, where each model represents a distinct equilibrium concept and implies a different mapping between observable market shares and firm margins. The posterior probability of a model is therefore not just a measure of statistical fit. It summarizes how well that conduct model reconciles the demand-side information in prices and shares with the supply-side information in margins.

This perspective also clarifies the relationship between Bayesian merger review and conventional calibration. Calibration implicitly places probability one on a single conduct model and probability zero on all alternatives. It is therefore a degenerate Bayesian procedure. Our framework replaces this all-or-nothing choice with explicit posterior uncertainty over conduct models and carries that uncertainty through to the merger counterfactual.

A second contribution is to show how identification changes with the available **information set**. We distinguish four canonical environments: (i) one market observed in one period, corresponding to the typical case-level merger review; (ii) one market observed over multiple periods; (iii) multiple markets observed in one period; and (iv) multiple markets observed over multiple periods. These environments form a nested sequence in which additional variation over time or across markets sharpens inference about demand, outside share, and conduct. Bayesian methods do not create identification where the likelihood is uninformative. They provide coherent inference when identification is weak, partial, or model-dependent.

We apply the framework to banking, with *United States v. Philadelphia National Bank* (1963) as the flagship

application. PNB is an ideal case because it is the canonical source of the structural presumption in merger law, it is historically important, and banking is one of the rare settings where shares, prices, and margin proxies can be observed. We use the broader Supreme Court banking merger setting only as external validation: the main question in this paper is how to conduct merger simulation when both the data and the conduct model are uncertain.

# 2 Structural Framework and Identification

We assume a standard Logit demand system where the market share $s_j$ of product $j$ is

$$s_j(\mathbf{p}) = \frac{\exp(\delta_j)}{1 + \sum_{k=1}^{J} \exp(\delta_k)},$$

with

$$\delta_j = x_j\beta - \alpha p_j + \xi_j.$$

Here $\alpha$ is the price sensitivity parameter and $s_0$ denotes the outside share.

## 2.1 Why Share-Consistent Demand?

We deliberately use a demand system that is consistent with share-based concentration metrics for three reasons rooted in the institutional reality of antitrust enforcement.

First, **legal doctrine operates on market shares**. From the *Philadelphia National Bank* structural presumption to the modern *Merger Guidelines*, agencies primarily screen and challenge mergers using share-based concentration measures such as HHI. A demand system that cannot map market structure into price effects is theoretically flexible but institutionally less useful. Among demand systems consistent with this share-centric paradigm, Logit and CES are the primary tractable options.

Second, **data constraints preclude unrestricted flexibility**. Random coefficients Logit allows richer substitution patterns, but it requires micro-data that is rarely available in the initial stages of merger review. In the absence of such data, flexible models often collapse back to calibration on shares, but with less transparency.

Third, our Bayesian approach **endogenizes market definition**. A common critique of Logit is its sensitivity to the outside good. Unlike standard calibration, which fixes $s_0$ ad hoc, we treat the outside share as an unknown parameter. This lets the model probabilistically size the market using the observed tension between prices, shares, and margins.

## 2.2 Margin-Share Relationships Across Conduct Models

The conduct models differ in how they map shares into markups. Under standard assumptions, each model implies a first-order condition of the form

$$p_j - c_j = \frac{1}{\alpha}\kappa_j^{(k)}(s_j, s_0),$$

where $\kappa_j^{(k)}(\cdot)$ is a model-specific markup function for conduct model $k$.

Equivalently, up to a common scale factor, the inverse margin relationship differs across conduct models:

**Bertrand (differentiated products):**

$$m_j^{-1} \propto 1 - s_j(1 - s_0).$$

**Second-score auction:**

$$m_j^{-1} \propto \frac{s_j(1 - s_0)}{-\log(1 - s_j(1 - s_0))}.$$

**Cournot (quantity competition):**

$$m_j^{-1} \propto \frac{s_0}{s_0 + s_j}.$$

**Monopolistic competition:**

$$m_j^{-1} \propto 1.$$

These mappings are the identification engine of the paper. The models are not merely alternative regressions or alternative priors. They impose different economic restrictions on the relationship between shares and margins.

### 2.3 Proposition 1: Structural Margin Hierarchy

**Proposition 1.** *For a fixed interior pair $(s_j, s_0)$ with $s_j > 0$ and $s_0 \in (0, 1)$, the candidate conduct models imply the following strict ordering of markup functions:*

$$\kappa_j^{Cournot}(s_j, s_0) > \kappa_j^{Bertrand}(s_j, s_0) > \kappa_j^{Auction}(s_j, s_0) > \kappa_j^{MonComp}(s_j, s_0).$$

*Proof.* Cournot competition scales the effect of own share by the outside share, yielding a markup factor proportional to $s_j/s_0$. Bertrand competition internalizes differentiated-product substitution and yields a markup factor proportional to $(1 - s_j(1 - s_0))^{-1} - 1$. Since $s_0 < 1$, the Cournot scaling is larger for interior shares. For the auction model, let $x = s_j(1 - s_0)$. The inequality $-\log(1 - x) > x$ for $x \in (0, 1)$ implies that the auction markup factor lies below the Bertrand markup factor. Monopolistic competition eliminates strategic interaction, leaving a constant own-effect benchmark. Combining these comparisons gives the stated ordering.

**Remark (Distinction from merger comparative statics).** The ordering in Proposition 1 is distinct from the comparative statics underlying merger thresholds. Prior work shows that different conduct models imply different counterfactual price effects for a given change in concentration. Proposition 1 instead establishes that these models also imply distinct relationships between observed market shares and margins in equilibrium. Because margins are observable, or can be proxied, these relationships generate testable restrictions that allow the data to discipline beliefs over conduct even in the absence of merger variation.

**Corollary 1.** *For a fixed dataset of shares, prices, and observed margins, the values of $\alpha$ required to rationalize those margins satisfy*

$$\alpha^{Cournot} > \alpha^{Bertrand} > \alpha^{Auction} > \alpha^{MonComp}.$$

The corollary explains how margins identify conduct. Each model prefers a different region of the parameter space to reconcile observed margins with observed shares. The demand side, through prices and shares, provides a separate anchor for $\alpha$. Posterior model probabilities are therefore disciplined by the tension between the demand-implied values of $\alpha$ and the supply-side markup mapping implied by each conduct model.

## 3 Information Sets and Identification

The strength of this identification mechanism depends on the data available to the analyst.

**One market, one period.** This is the canonical merger-review environment. Shares and prices identify relative mean utilities, but without margins they do not separately identify $\alpha$, unobserved quality, the outside share, or conduct. Observed margin proxies add discipline because each conduct model implies a different mapping from $(s_j, s_0, \alpha)$ to margins. Even then, inference remains weak and relies heavily on functional form, priors, and cross-firm variation within the market.

**One market, multiple periods.** Time-series variation helps separate persistent firm quality from price sensitivity and allows margins to discipline the same market under changing conditions. This improves inference about $\alpha$ and $s_0$, but the analysis remains tied to a single market.

**Multiple markets, one period.** Cross-market variation makes it possible to learn whether the same conduct model explains different market structures. This setting begins to identify the distribution of conduct across markets, rather than only the conduct model for one case.

**Multiple markets, multiple periods.** The full hierarchical setting combines cross-market and time-series variation. It allows partial pooling of parameters such as $\alpha$ and $s_0$, separates case-specific noise from persistent conduct patterns, and provides the most informative prior for future case-level merger review.

This information-set ladder is central to the interpretation of Bayesian merger analysis. Case-level review often occurs in the weakest environment, while broader empirical work can use richer data to learn priors over conduct and demand that improve future adjudication. These cases form a nested identification ladder. The same structural model is estimated in each environment, but the extent to which the likelihood disciplines parameters and conduct varies sharply with the available variation. This allows us to separate what is learned from the data from what is imposed by structure or priors.

# 4 Bayesian Framework

The theoretical results above establish that conduct models imply distinct mappings between shares and margins. We now embed those mappings in a Bayesian likelihood.

## 4.1 Likelihood

Let $D = \{s_j, p_j, m_j\}_{j=1}^{J}$ denote observed shares, prices, and margins or margin proxies in a market. For each conduct model $M_k \in \{\text{Bertrand}, \text{Cournot}, \text{Auction}, \text{MonComp}\}$, the structural model implies a joint demand-supply system:

$$s_j = \exp(\delta_j + \log s_0) + \varepsilon_j^{(s)}, \quad \delta_j = x_j\beta - \alpha p_j + \xi_j,$$

and

$$m_j^{-1} = \frac{1}{\alpha}\kappa_j^{(k)}(s_j, s_0) + \varepsilon_j^{(m)}.$$

Because $\kappa_j^{(k)}(\cdot)$ differs across conduct models as characterized in Proposition 1, the margin equation imposes model-specific restrictions linking observed shares to margins. These restrictions generate differences in the joint likelihood across models.

We allow the share and margin disturbances to be correlated:

$$\begin{pmatrix} \varepsilon_j^{(s)} \\ \varepsilon_j^{(m)} \end{pmatrix} \sim \mathcal{N}\left( \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_s^2 & \rho\sigma_s\sigma_m \\ \rho\sigma_s\sigma_m & \sigma_m^2 \end{pmatrix} \right).$$

The likelihood under model $M_k$ is therefore the joint fit of shares and margins:

$$P(D \mid \theta_k, M_k) = \prod_{j=1}^{J} f_s(s_j \mid \theta_k, M_k) f_m(m_j \mid s_j, \theta_k, M_k),$$

where $\theta_k$ includes $\alpha$, $s_0$, demand shifters, latent quality, variance parameters, and the correlation parameter $\rho$.

## 4.2 Priors and Hierarchical Structure

We use weakly informative priors for structural parameters. The price sensitivity $\alpha$ is restricted to be positive, the outside share $s_0$ is restricted to the unit interval, latent quality terms are regularized hierarchically, and the variance parameters are constrained to be positive. In multi-market or multi-period settings, parameters such as $\alpha$ and $s_0$ may vary hierarchically across events, markets, or years, allowing richer information sets to discipline case-level inference through partial pooling.

### 4.3 Model Comparison and Averaging

For each conduct model, we compute the marginal likelihood

$$P(D \mid M_k) = \int P(D \mid \theta_k, M_k) P(\theta_k \mid M_k) d\theta_k.$$

Posterior model probabilities are then

$$P(M_k \mid D) = \frac{P(D \mid M_k) P(M_k)}{\sum_\ell P(D \mid M_\ell) P(M_\ell)}.$$

We typically use uniform prior probabilities over conduct models.

Predicted merger effects are averaged over conduct models:

$$E[\Delta p \mid D] = \sum_k E[\Delta p \mid D, M_k] P(M_k \mid D).$$

The object being averaged is not a reduced-form forecast. It is a merger counterfactual solved under a distinct equilibrium model. The resulting BMA estimate is therefore a decision-theoretic object: the expected merger effect after accounting for uncertainty over both structural parameters and firm conduct.

### 4.4 Implementation

The models are estimated in Stan using the HMC NUTS sampler. We approximate model evidence using marginal likelihood or predictive-fit criteria, depending on the application. The key empirical comparison is the joint ability of each conduct model to reconcile observed shares, prices, and margins.

## 5 Application: The Banking Industry

We apply the framework to the U.S. banking industry, a sector where merger review, market definition, and concentration screens have long been central policy questions. Banking is especially useful for this paper because it combines institutional importance with unusually rich information on shares, prices, and margin proxies. That combination lets us study exactly the object emphasized above: how conduct models reconcile demand-side data with supply-side margins.

The industry is also theoretically ambiguous. Banks may compete over deposit rates in a differentiated-products Bertrand environment, over quantities of loanable funds in a Cournot-like environment, or through relationship-based bidding that resembles auction competition. Because each of these conduct assumptions can be economically plausible, banking is a natural setting in which to treat conduct as an object of inference rather than as a maintained assumption.

This application has three advantages. First, the legal doctrine of the structural presumption originates in the Supreme Court's *Philadelphia National Bank* decision, making banking directly relevant to merger policy. Second, banking accounting data allow construction of margin proxies that are rarely available in other industries. Third, local banking markets generate cross-sectional variation that can be used to discipline structural parameters and model probabilities.

## 6 Case Study: United States v. Philadelphia National Bank (1963)

To demonstrate the practical utility of our Bayesian framework, we apply it to the data from the landmark *United States v. Philadelphia National Bank* (1963) case. This case established the "structural presumption" in antitrust law, yet the nature of competition in this market remains a subject of theoretical interest.

## 6.1 Data Overview

We utilize bank-level data from the year **1960**, comprising `39` observations for the relevant banking market. By focusing on a single cross-section, we establish a baseline for structural identification before extending to the full panel. The observed data reveals significant heterogeneity in market shares and operating margins.

Table 1: Summary Statistics: PNB Case Data (Pooled 1954-1960)

| Variable | Mean | SD | Min | Max |
|---|---|---|---|---|
| Margin | 1.445 | 0.457 | 0.214 | 2.806 |
| shareIn | 0.030 | 0.057 | 0.000 | 0.234 |
| rate_deposits | 2.597 | 0.514 | 0.561 | 3.603 |

## 6.2 Bayesian Model Comparison and Averaging

We evaluate four structural models of competition: **Bertrand** (differentiated price setting), **Auction** (bidding for relationships), **Cournot** (quantity competition), and **Monopolistic Competition** (independent pricing).

To ensure a fair comparison, we employ **model-specific data-calibrated priors**. Since the relationship between margins and shares differs fundamentally across models (e.g., Cournot implies higher margins than Bertrand for the same share), imposing a common $\alpha$ prior can bias model selection. Instead, we derive weakly informative priors for each model based on the observed data extremes, allowing each theory to be tested on its own terms.

We compare these models using model evidence and predictive fit. The results for the PNB market are summarized below.

Table 2: Bayesian Model Comparison (Model-Specific Priors)

| Model | LogML | Divergences | BF vs Best | Post. Prob. |
|---|---|---|---|---|
| Cournot | 103.486 | 9 | 1.000 | 0.984 |
| Bertrand | 98.832 | 29 | 0.010 | 0.009 |
| Auction | 98.170 | 12 | 0.005 | 0.005 |
| MonCom | 97.338 | 11 | 0.002 | 0.002 |

The data favor the **Cournot** model by log marginal likelihood, indicating that its share-margin mapping best reconciles the observed 1960 PNB data under the specified priors. The **Bertrand** and **Auction** models remain economically plausible, but they require different regions of the parameter space to match the same margins and therefore receive lower posterior weight in this single-year information set.

**Structural interpretation.** The structural parameter $\alpha$ is estimated to be approximately **0.37-0.44** across models. This range is consistent with the identification logic above: different conduct models imply different markup mappings, so they rationalize the same observed margins with different demand curvature. The estimated outside share is small across models, consistent with the narrow market definition used in the case.

**Convergence diagnostics.** The sampler produces remaining divergences across models, especially in the most weakly identified specifications. We therefore interpret the single-year PNB estimates as an intentionally demanding case-level exercise rather than as a fully saturated panel estimate.

### 6.2.1 Bayesian Model Averaging (BMA)

To account for model uncertainty, we compute the Bayesian Model Averaged (BMA) parameters.

Table 3: Parameter Estimates by Model (Model-Specific Priors)

| Parameter | Prior | Bertrand | Auction | Cournot | MonCom |
|---|---|---|---|---|---|
| Alpha | 1.00 (0.27) | 0.433 (0.047) | 0.435 (0.046) | 0.516 (0.069) | 0.438 (0.045) |
| S0 | 0.05 (1.20) | 0.011 (0.001) | 0.011 (0.001) | 0.011 (0.001) | 0.011 (0.001) |
| Sigma_Share_Abs | 0.000 (0.010) | 0.002 (0.001) | 0.002 (0.001) | 0.002 (0.001) | 0.002 (0.001) |
| Sigma_Margin | 0.000 (1.00) | 0.687 (0.082) | 0.695 (0.083) | 0.671 (0.082) | 0.702 (0.086) |
| Rho | 0.000 (0.50) | 0.000 (0.000) | 0.000 (0.000) | 0.000 (0.000) | 0.000 (0.000) |
| Cutoff_Share | 0.03 (0.02) | NA (NA) | NA (NA) | NA (NA) | NA (NA) |
| Sigma_B_Strat | 0.0 (5.0) | 1.929 (0.232) | 1.927 (0.236) | 1.942 (0.229) | 1.926 (0.233) |
| Sigma_B_Fringe | 0.0 (0.2) | 0.000 (0.000) | 0.000 (0.000) | 0.000 (0.000) | 0.000 (0.000) |

#### 6.2.2 Key Parameter Estimates

The table above shows parameter estimates across all four models. Note:

- **Alpha** (price sensitivity): Estimates range from 0.37 (Bertrand) to 0.44 (Cournot).
- **S0** (outside share): Effectively **0.6-0.7%** across all models, validating the strict market definition.
- **Sigma_Margin**: ~0.49-0.66, indicating reasonable fit to the pricing equations.

Table 4: Simulated Merger Price Effects: Bayesian vs. Calibration (1960 Snapshot)

| Model | Bayesian Effect (%) | Calibration Effect (%) |
|---|---|---|
| Cournot | -118.32 | -35.62 |
| Bertrand | -12.79 | -26.80 |
| Auction | -6.64 | -4.09 |
| MonCom | 0.00 | 0.00 |

The difference between Bayesian and single-model predictions is economically meaningful. In the current 1960 PNB snapshot, standard Bertrand calibration predicts a mean effect of approximately **-26.8%**, while the posterior-weighted Bayesian model average is approximately **-116.5%**. The gap arises because the observed share-margin relationship places substantial posterior weight on Cournot conduct. This illustrates the main policy point of the paper: conditioning on a single conduct model can materially change the counterfactual conclusion even in a canonical case such as PNB.

#### 6.2.3 Remark: Neutrality of Monopolistic Competition

Formally, the zero price effect in the Monopolistic Competition model arises from the assumption of strategic independence. Consider the profit maximization problem for firm $j$:

$$\max_{p_j} \pi_j = (p_j - c_j) q_j(p_j)$$

The first-order condition (FOC) is:

$$\frac{\partial \pi_j}{\partial p_j} = q_j + (p_j - c_j) \frac{\partial q_j}{\partial p_j} = 0$$

Under the Monopolistic Competition assumption, firms perceive the cross-price elasticities as zero (i.e., $\frac{\partial q_k}{\partial p_j} = 0$ for $k \neq j$).

Following a merger between firms $j$ and $k$, the merged entity maximizes joint profit $\pi_M = \pi_j + \pi_k$. The FOC with respect to $p_j$ becomes:

$$\frac{\partial \pi_M}{\partial p_j} = \underbrace{\left[q_j + (p_j - c_j)\frac{\partial q_j}{\partial p_j}\right]}_{\text{Own-Effect}} + \underbrace{(p_k - c_k)\frac{\partial q_k}{\partial p_j}}_{\text{Cross-Effect (Merger)}} = 0$$

Because the model assumes $\frac{\partial q_k}{\partial p_j} = 0$, the "Cross-Effect" term vanishes. The post-merger FOC is identical to the pre-merger condition, implying that ownership consolidation creates no incentive to raise prices.

# 7 External Validation: Historic Banking Mergers

PNB is the flagship application because it is the canonical antitrust banking case and the origin of the structural presumption. We also use other historic Supreme Court banking mergers as a limited external validation exercise. The purpose of this extension is not to relitigate those cases or to develop a separate legal account of Supreme Court merger doctrine. The purpose is to ask whether the same Bayesian conduct framework can be estimated in related banking markets with different sizes and concentration levels.

In this broader exercise, the cutoff/fringe specification separates banks that appear to participate strategically from the long tail of very small institutions. This matters in fragmented banking markets, where many small banks contribute observations but may not discipline strategic conduct in the same way as larger competitors. The multi-case setting is therefore best interpreted as movement along the information-set ladder: from a single PNB cross-section toward a richer panel that can learn more about the distribution of conduct across markets.

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## **Supreme Court analysis results pending...** (Data not yet loaded)
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This extension provides a check on external relevance. If the framework performs similarly across related banking markets, then the PNB results are less likely to be an artifact of a single historical case. The central contribution remains the Bayesian treatment of conduct uncertainty in merger simulation.

This extension should therefore be interpreted as movement along the information-set ladder, rather than as a separate exercise in legal interpretation.

# 8 Conclusion

We have presented a Bayesian framework for merger review that explicitly accounts for uncertainty over firm conduct. The framework shifts the analysis from selecting a single "best" model to estimating posterior beliefs over economically distinct conduct models and carrying those beliefs into merger counterfactuals.

More broadly, the framework suggests that merger analysis should be viewed as inference under model uncertainty, rather than as simulation under a maintained conduct assumption.

## 8.1 Policy Implications

Our results suggest that relying on a single structural model, often Bertrand, can produce overconfident merger predictions when conduct is uncertain. The Bayesian approach does not eliminate the need for judgment or identifying assumptions, especially in one-market, one-period cases. It does make those assumptions explicit, quantifies how much the data support alternative conduct models, and reports merger effects that incorporate model uncertainty.

## 8.2 Future Directions

While this paper focuses on case-specific adjudication, a natural extension is to use richer multi-market and multi-period data to construct priors for future merger reviews. Such priors could inform conduct-robust

screening thresholds and reduce the instability of rules based only on concentration measures. Extending the framework to more flexible demand systems, such as random coefficients Logit, is another important direction.

# 9 Appendix A: Data Quality and Outlier Analysis

During the multi-year analysis (1960-1965), we identified a critical data anomaly that necessitated a surgical filtering step. This appendix documents the issue and its resolution.

## 9.1 The 66-Sigma Outlier

In the raw panel data, one observation exhibited an extreme deviation from the sample distribution: - **Observation:** Union NB&TC Souderton (1962) - **Market Share:** 0.37% (Fringe) - **Inverse Margin: 32.8** (implied margin ~3%)

The rest of the dataset ($N = 201$) is highly concentrated, with a mean Inverse Margin of **0.80** and a standard deviation of **0.48**. The outlier represents a Z-score of approximately **66.7**:

$$Z = \frac{32.8 - 0.80}{0.48} \approx 66.7$$

## 9.2 Impact on Structural Estimation

Including this single point in a standard Normal likelihood framework forces the model to either: 1. Assign near-zero probability to the data (sampler failure), or 2. Inflate the estimated error variance ($\sigma$) to accommodate the outlier.

In initial test runs, the model inflated $\sigma_{margin}$ to **2.28**, rendering the structural parameter $\alpha$ unidentifiable (posterior collapsed to ~0.02).

## 9.3 Resolution

To preserve structural integrity, we applied a **Surgical Filter**: > `filter(margin >= 0.1)` (Equivalent to Inverse Margin $\leq 10$)

This filter removes exactly **1 observation** (the outlier) while retaining **201 observations**, including 133 other small banks. This approach ensures that the structural parameters are estimated from the representative sample of market behavior rather than being dominated by a single data error.

# 10 References

Panhans, Matthew T., and Charles Taragin. 2023. "Consequences of Model Choice in Predicting Horizontal Merger Effects." *International Journal of Industrial Organization* 89: 102986. https://doi.org/10.1016/j.ijindorg.2023.102986.